

C870

Human Anatomy and Physiology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher who coaches the school's track team uses the leg of a life-size skeleton, bought in 2018, to review terms. Which statement is correct in anatomical position?

- A. The knee is medial to the ankle.
- B. The knee is anterior to the ankle.
- C. The knee is distal to the ankle.
- D. The knee is proximal to the ankle.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which structure is distal to the elbow?

- A. The shoulder
- B. The sternum
- C. The wrist
- D. The humerus

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

In a lab with 12 microscopes donated by a local clinic, students view a stained blood smear. It shows red cells, white cells, and platelets in a fluid matrix. Which tissue type is blood?

- A. Epithelial tissue
- B. Muscle tissue, since the heart pumps it
- C. Connective tissue
- D. It is not a tissue because it is a liquid.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

After a 5-mile run on a humid 32°C afternoon, a runner's body temperature rises and she begins to sweat, which cools her. She drank 500 mL of water beforehand. What kind of control is this?

- A. Negative feedback
- B. Positive feedback, because sweating increases as temperature increases
- C. Positive feedback, because it helps her
- D. No feedback; sweating is voluntary.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which process is an example of positive feedback?

- A. Oxytocin increasing contractions during childbirth, which release more oxytocin
- B. Any response that helps the body
- C. Shivering when body temperature drops
- D. Insulin release after a meal lowering blood glucose

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Students dissect a sheep heart ordered from a supplier that ships on ice. The heart weighs about 250 g. Which vessels carry oxygen-rich blood into the left atrium?

- A. The pulmonary arteries
- B. The pulmonary veins
- C. The aorta
- D. The venae cavae

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Blood in the pulmonary artery is best described as:

- A. Oxygen-rich blood returning from the lungs
- B. Oxygen-poor blood on its way to the lungs
- C. Oxygen-rich, because arteries always carry oxygenated blood
- D. Blood headed to the body tissues

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A swim coach who has trained athletes for 20 years asks how oxygen moves from the alveoli into the blood. Alveolar walls are one cell thick, and the lungs hold roughly 300 million alveoli. What is the mechanism?

- A.** Passive diffusion from higher to lower partial pressure
- B.** Osmosis
- C.** Active transport by pumps in the alveolar wall
- D.** Diffusion of oxygen from the blood into the alveoli

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A student giving a speech to an audience of 150 feels her heart pound. She ate lunch an hour earlier. Her sympathetic nervous system is active. Which response is expected?

- A.** Digestion speeds up to process her lunch.
- B.** Every organ speeds up.
- C.** Heart rate slows and airways narrow.
- D.** Pupils dilate and digestion slows.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Two hours after a large pasta dinner at a restaurant that seats 80, a person's blood glucose rises. The meal cost \$18. Which hormone does the pancreas release, and what does it do?

- A.** Glucagon, which lowers blood glucose
- B.** Insulin, which moves glucose into cells and lowers blood glucose
- C.** Glucagon, which stores glucose in the liver
- D.** Cortisol, the main hormone for lowering glucose

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A hiker on a 14-mile trail, who carried a new backpack, runs out of water and becomes dehydrated. ADH secretion increases. What happens to the hiker's urine?

- A.** Unchanged, since the kidneys filter at a constant rate
- B.** More glucose appears in it.
- C.** Larger volume and more dilute
- D.** Smaller volume and more concentrated

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A substance blocks the release of ADH. What change in urine is expected?

- A.** Smaller volume of concentrated urine
- B.** More sodium is reabsorbed, so urine volume drops.
- C.** Larger volume of dilute urine
- D.** No urine is made.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A teacher ~~who coaches the school's track team~~ uses the leg of a life-size skeleton, ~~bought in 2018~~, to review terms. Which statement is correct in anatomical position?

Target word: correct directional term **Decides:** Proximal = closer to the limb's attachment at the trunk.

A	Adjacent category	Medial and lateral describe position relative to the midline, not along a limb.
B	Confusion set	Anterior/posterior describe front and back, not position along a limb.
C	Inverted operation	Reverses proximal and distal.
D	Correct	The knee is closer to the hip, where the leg attaches.

2. Answer C Concept 1 • transfer

Which structure is distal to the elbow?

Target word: structure from a term **Decides:** Distal = farther from the shoulder attachment.

A	Inverted operation	Proximal to the elbow.
B	Adjacent category	Part of the trunk, not the limb.
C	Correct	Farther along the arm from the shoulder.
D	Confusion set	The upper-arm bone lies proximal to the elbow.

3. Answer C Concept 2

In a lab ~~with 12 microscopes donated by a local clinic~~, students view a stained blood smear. It shows red cells, white cells, and platelets in a fluid matrix. Which tissue type is blood?

Target word: tissue type of blood **Decides:** Cells in a fluid extracellular matrix (plasma).

A	Confusion set	Epithelium lines surfaces and has little matrix.
B	Adjacent category	The heart is muscle; blood is not.
C	Correct	Blood is a fluid connective tissue.
D	Absolute language	Tissues can have fluid matrices.

4. Answer A Concept 3

After a 5-mile run on a humid 32°C afternoon, a runner's body temperature rises and she begins to sweat, which cools her. ~~She drank 500 mL of water beforehand.~~ What kind of control is this?

Target word: type of feedback **Decides:** Sweating reverses the temperature rise.

A	Correct	The response returns temperature toward the set point.
B	Inverted operation	The response opposes the change in temperature.
C	Confusion set	"Positive" means amplifying, not beneficial.
D	Adjacent category	Sweating is an automatic, regulated response.

5. Answer A Concept 3 • transfer

Which process is an example of positive feedback?

Target word: reverse: pick the amplifying loop **Decides:** The response increases the original stimulus.

A	Correct	The response amplifies the stimulus until birth.
B	Absolute language	Most helpful responses are negative feedback.
C	Confusion set	Negative feedback that restores temperature.
D	Inverted operation	Negative feedback: it reverses the rise.

6. Answer B Concept 4

Students dissect a sheep heart ~~ordered from a supplier that ships on ice. The heart weighs about 250 g.~~ Which vessels carry oxygen-rich blood into the left atrium?

Target word: vessels into left atrium **Decides:** Blood returns from the lungs through pulmonary veins.

A	Inverted operation	They carry oxygen-poor blood away from the right ventricle.
B	Correct	They return oxygenated blood from the lungs to the left atrium.
C	Adjacent category	The aorta leaves the left ventricle; it does not enter the atrium.
D	Confusion set	They bring oxygen-poor blood to the right atrium.

7. Answer B Concept 4 • transfer

Blood in the pulmonary artery is best described as:

Target word: oxygen content of pulmonary artery blood **Decides:** It leaves the right ventricle toward the lungs.

A	Inverted operation	Describes the pulmonary veins.
B	Correct	Pulmonary arteries carry deoxygenated blood.
C	Absolute language	Arteries are defined by direction (away from the heart), not oxygen.
D	Adjacent category	That is the aorta's systemic path.

8. Answer A Concept 5

A swim coach ~~who has trained athletes for 20 years~~ asks how oxygen moves from the alveoli into the blood. Alveolar walls are one cell thick, ~~and the lungs hold roughly 300 million alveoli.~~ What is the mechanism?

Target word: mechanism of O₂ movement **Decides:** Higher O₂ partial pressure in alveoli than in capillary blood.

A	Correct	No energy needed; gases follow their gradients.
B	Confusion set	Osmosis is water movement, not gas movement.
C	Adjacent category	Gas exchange uses no ATP-driven pumps.
D	Inverted operation	That is the direction CO ₂ moves.

9. Answer D Concept 6

A student giving a speech ~~to an audience of 150~~ feels her heart pound. ~~She ate lunch an hour earlier.~~ Her sympathetic nervous system is active. Which response is expected?

Target word: sympathetic effect **Decides:** Fight-or-flight response.

A	Camouflage	The lunch detail does not override sympathetic slowing of digestion.
B	Absolute language	The sympathetic system slows some functions, such as digestion.
C	Inverted operation	Those are parasympathetic effects.
D	Correct	Sympathetic activity diverts resources from digestion.

10. Answer B Concept 7

Two hours after a large pasta dinner ~~at a restaurant that seats 80~~, a person's blood glucose rises. ~~The meal cost \$18.~~ Which hormone does the pancreas release, and what does it do?

Target word: hormone and effect after a meal **Decides:** High glucose triggers insulin.

A	Inverted operation	Glucagon raises glucose.
B	Correct	Beta cells respond to high glucose.
C	Confusion set	Swaps the role of insulin.
D	Adjacent category	Cortisol tends to raise glucose; it is not the pancreatic response.

11. Answer D Concept 8

A hiker ~~on a 14-mile trail, who carried a new backpack~~, runs out of water and becomes dehydrated. ADH secretion increases. What happens to the hiker's urine?

Target word: effect of ADH **Decides:** ADH increases water reabsorption.

A	Absolute language	Filtration may be steady, but reabsorption changes.
B	Adjacent category	Glucose in urine relates to insulin and blood glucose, not ADH.
C	Inverted operation	That is the effect of low ADH.
D	Correct	More water is returned to the blood.

12. Answer C Concept 8 · transfer

A substance blocks the release of ADH. What change in urine is expected?

Target word: reverse: effect of low ADH **Decides:** Less water reabsorbed in collecting ducts.

A	Inverted operation	That is the effect of high ADH.
B	Confusion set	Sodium reabsorption is driven mainly by aldosterone, not ADH.
C	Correct	Without ADH, more water stays in the filtrate.
D	Absolute language	Urine still forms; only its concentration changes.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	0	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Directional terms	Terms are relative to anatomical position: proximal/distal refer to closeness to a limb's attachment; anterior/posterior to front/back; superior/inferior to above/below.	Using superior/inferior for limb positions, or forgetting anatomical position.	Two structures on the same limb or body region.
2	The four tissue types	All tissues are epithelial, connective, muscle, or nervous; blood, bone, fat, and tendons are connective tissue.	Classifying blood as its own tissue type or as muscle.	A structure and a question about tissue type.
3	Negative vs. positive feedback	Negative feedback reverses a change to restore a set point; positive feedback amplifies a change until an endpoint is reached.	Calling any helpful response positive feedback.	Whether the response pushes the variable back or further in the same direction.
4	Pulmonary vs. systemic circulation	The right heart pumps oxygen-poor blood to the lungs through pulmonary arteries; pulmonary veins return oxygen-rich blood to the left atrium.	Assuming all arteries carry oxygen-rich blood.	"Pulmonary artery" or "pulmonary vein."
5	Gas exchange by diffusion	O ₂ and CO ₂ move between alveoli and capillaries by passive diffusion down partial-pressure gradients.	Thinking gas exchange uses active transport or that CO ₂ moves into the blood at the lungs.	Alveoli, capillaries, and the direction a gas moves.
6	Sympathetic vs. parasympathetic	Sympathetic (fight or flight) raises heart rate, dilates pupils and airways, and slows digestion; parasympathetic (rest and digest) does the reverse.	Thinking the sympathetic system speeds up everything, including digestion.	Stress or rest scenario and a named organ response.
7	Insulin vs. glucagon	Insulin (beta cells) lowers blood glucose by moving it into cells; glucagon (alpha cells) raises it by releasing stored glucose from the liver.	Mixing up which hormone raises and which lowers glucose.	A meal or a fast and a question about pancreatic hormones.
8	ADH and water balance	ADH increases water reabsorption in the collecting ducts, producing a smaller volume of concentrated urine.	Thinking more ADH means more urine.	Dehydration, alcohol, or urine volume and concentration.

Course scope (public description): This course examines the structures and functions of the human body and covers anatomical terminology, cells and tissues, and organ systems. Students will study the healthy state of the organ systems of the human body, including the digestive, skeletal, sensory, respiratory, reproductive, nervous, muscular, cardiovascular, lymphatic, integumentary, endocrine, and renal systems. There are no prerequisites for this course.